

KONGSVINGER, Norway

WMO: 014680

Lat: 60.1903N

Lon: 12.0067E

Elev: 148

StdP: 99.56

Time Zone: 1.00 (EUC)

Period: 06-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-21.3	-17.8	-23.3	0.5	-21.3	-19.6	0.7	-17.6	7.2	4.2	6.1	2.7	1.3	100	0.570

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.5	27.0	16.9	24.9	16.0	22.8	15.3	18.2	23.9	17.3	22.5	16.3	20.9	2.0	300

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
16.2	11.7	19.8	15.3	11.0	18.9	14.4	10.4	18.1	52.2	23.8	49.1	22.6	46.2	21.0	22.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5.6	4.9	4.2	-23.8	29.5	3.4	2.3	-26.2	31.1	-28.3	32.4	-30.2	33.7	-32.7	35.4
			-24.5	19.6	3.7	1.3	-27.1	20.6	-29.3	21.3	-31.4	22.1	-34.0	23.1
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	5.7	-5.9	-3.7	0.3	5.2	10.3	14.4	17.0	15.1	11.3	5.3	0.9	-2.8
	DBStd	8.66	6.46	5.25	4.33	3.43	3.88	2.95	2.89	2.28	2.86	3.46	4.25	5.71
	HDD10.0	2210	492	384	301	148	43	2	0	0	19	150	274	397
	HDD18.3	4651	750	618	559	393	249	122	64	103	211	403	524	655
	CDD10.0	630	0	0	0	5	53	135	216	158	58	5	0	0
	CDD18.3	29	0	0	0	0	1	5	21	3	0	0	0	0
	CDH23.3	357	0	0	0	0	32	88	219	18	0	0	0	0
	CDH26.7	64	0	0	0	0	2	12	51	0	0	0	0	0
Wind	WSAvg	1.8	1.8	1.8	1.9	2.0	2.0	1.9	1.8	1.7	1.7	1.7	1.8	1.9
Precipitation	PrecAvg	724	48	36	32	48	53	80	80	88	64	75	69	52
	PrecMax	1028	83	70	93	97	136	133	120	149	112	196	189	98
	PrecMin	615	10	9	1	3	8	19	24	16	22	23	23	9
	PrecStd	105	23	19	25	25	35	33	29	39	28	40	44	28
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.7	7.1	14.7	19.7	26.2	28.5	30.3	25.4	21.8	15.8	11.1	9.4
		MCWB	5.0	4.1	7.9	10.6	14.9	16.2	18.3	17.1	15.5	12.2	9.3	7.8
	2%	DB	4.9	5.2	11.1	17.6	23.9	25.9	28.0	23.3	19.4	13.6	9.5	7.6
		MCWB	3.6	3.2	5.7	10.0	13.8	15.3	17.7	16.2	14.1	10.9	8.1	6.2
	5%	DB	3.5	4.0	8.1	15.1	21.1	23.7	25.9	21.5	17.6	12.1	8.2	6.0
		MCWB	2.5	2.6	3.9	8.5	12.9	14.8	17.4	15.3	13.3	9.8	6.9	4.9
	10%	DB	1.8	2.6	6.1	12.6	18.2	21.1	23.5	19.9	16.2	10.7	6.7	4.5
		MCWB	0.9	1.5	3.1	7.1	11.6	13.4	16.3	14.7	12.5	8.8	5.7	3.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.4	5.0	8.4	11.7	16.6	17.7	20.2	18.6	16.6	13.2	9.7	8.0
		MCDB	6.4	6.2	14.0	17.3	22.5	23.2	26.7	21.9	18.9	14.8	10.7	9.3
	2%	WB	3.9	3.7	6.1	10.3	15.0	16.6	18.9	17.4	15.2	11.7	8.2	6.2
		MCDB	4.7	4.8	10.3	16.1	21.5	23.1	25.2	21.1	17.7	13.0	9.3	7.3
	5%	WB	2.5	2.7	4.6	9.1	13.6	15.5	17.9	16.5	14.1	10.3	7.0	4.9
		MCDB	3.4	3.7	7.2	14.3	19.4	22.0	24.4	20.1	17.0	11.7	8.0	5.9
	10%	WB	1.1	1.6	3.5	7.7	12.3	14.4	17.0	15.6	13.0	9.0	5.8	3.3
		MCDB	1.7	2.5	5.8	12.0	17.3	20.1	22.4	18.7	15.5	10.4	6.7	4.3
Mean Daily Temperature Range	5% DB	MDBR	5.8	6.7	9.2	10.8	10.4	10.0	9.5	8.6	8.5	6.8	4.9	5.2
		MCDBR	5.6	7.5	12.5	15.3	16.0	14.7	13.7	11.2	10.8	8.5	6.7	6.3
		MCWBR	5.1	5.7	7.7	8.3	7.2	5.9	5.5	5.5	6.2	5.8	5.7	5.7
	5% WB	MCDBR	5.5	6.4	11.1	14.0	13.2	12.3	12.3	9.4	8.9	6.9	6.3	6.1
		MCWBR	5.0	5.1	7.1	7.8	6.5	5.7	5.3	5.3	5.5	5.3	5.6	5.7
Clear-Sky Solar Irradiance	taub	0.255	0.298	0.324	0.365	0.364	0.354	0.374	0.366	0.334	0.321	0.275	0.228	
	taud	2.202	2.283	2.334	2.335	2.407	2.457	2.415	2.436	2.514	2.460	2.285	2.112	
	Ebn at Noon	525	703	806	832	863	878	847	826	797	680	498	397	
	Edh at Noon	40	66	86	104	104	100	103	93	73	56	37	27	
All-Sky Solar Radiation	RadAvg	0.28	0.91	2.40	3.79	4.81	5.50	4.92	3.86	2.56	1.14	0.37	0.18	
	RadStd	0.05	0.17	0.26	0.49	0.48	0.39	0.66	0.36	0.31	0.22	0.10	0.02	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (59 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page